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Studierichting: Informatica

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$$\begin{aligned}\int \frac{x^2 + 6}{x^3 - 2x - 4} dx &= \int \frac{x^2 + 6}{(x - 2)(x^2 + 2x + 2)} dx \\&\rightarrow \frac{x^2 + 6}{(x - 2)(x^2 + 2x + 2)} = \frac{A}{x - 2} + \frac{Bx + C}{x^2 + 2x + 2} \\&\Rightarrow A = \frac{4 + 6}{4 + 4 + 2} = 1 \\&\Rightarrow B(-1 + i) + C = \frac{(-1 + i)^2 + 6}{-1 + i - 2} = -2 + 0i \Rightarrow B = 0, C = -2 \\&= \int \frac{1}{x - 2} d(x - 2) - 2 \int \frac{1}{(x + 1)^2 + 1} d(x + 1) \\&= \ln |x - 2| - 2 \operatorname{Bgtan}(x + 1) + c\end{aligned}$$

References